One Model, Many Geometries: Spectral Dynamics of Multilingual LLMs

Mattia Ferrarini | 407144 | mattia.ferrarini@epfl.ch
João Pinto | 407597 | joao.azevedopinto@epfl.ch
Giacomo Porpiglia | 405044 | giacomo.porpiglia@epfl.ch
Pedro Pinto | 408828 | pedro.azevedopinto@epfl.ch
JPMG

Abstract

We explore the geometric evolution of multilingual LLM representations during pretraining, tracking per-language RankMe across 50+ checkpoints for two multilingual models and 10–15 diverse languages. We ask whether geometric phases align across languages, what drives any differences, and whether divergence predicts downstream performance.

1 Introduction

While loss decreases monotonically during LLM pretraining, Li et al. (2025) show that representation geometry evolves in three non-monotonic phases. However, their analysis is limited to English models, and multilingual training dynamics have largely been studied using only feature-level methods or sparse checkpoints (Blevins et al., 2022; Wang et al., 2024; Chang et al., 2022).

We hypothesize that the macro-level evolution of per-language representations predicts multilingual capabilities. To test this, we track per-language spectral measures across 50+ checkpoints for two multilingual models, asking whether (1) languages share geometric phases, (2) differences are explained by language properties or data distribution, and (3) these differences predict downstream performance and cross-lingual transfer.

2 Approach

Models. We use two open-source 8B multilingual LLMs: `Apertus-8B` (Apertus et al., 2025) (71 checkpoints), and `FuxiTranyu-8B` (Sun et al., 2024) (58 checkpoints).

RankMe. Our primary metric for analyzing geometry evolution is RankMe (Garrido et al., 2023), a measure of effective rank also used by Li et al. (2025), whose definition we adopt. For a layer L with hidden dimension D, we collect $N \gg D$ activations (e.g., the last token from N sequences) into a matrix X. We then compute the centered covariance matrix Σ, and normalize its eigenvalues λ to yield a probability distribution $p_k = \lambda_k / \sum_j \lambda_j$. RankMe is the exponentiated entropy of p: $\text{RankMe}(X) = \exp(-\sum_k p_k \ln p_k)$. We compute RankMe for a fixed L across all checkpoints of both models for 10–15 diverse languages.

Datasets and benchmarks. We compute RankMe on language-specific subsets of `FineWeb2` (Penedo et al., 2025). We use `XCOPA` (Ponti et al., 2020; Roemmele et al., 2011) and `m-MMLU` (Hendrycks et al., 2020) to evaluate downstream multilingual abilities, and `ECLeKTic` (Goldman et al., 2025) to evaluate cross-lingual transfer. We also test zero-shot transfer using pairs of language-specific subsets from `XNLI` (Conneau et al., 2018).

3 Project Timeline

In **Phase 1** (May 3–May 24), we perform representation extraction and geometry analysis, implementing the pipeline to extract RankMe metrics from our checkpoints and mapping the geometric evolution of all selected languages. In **Phase 2** (May 25–June 7), we evaluate capabilities and synthesize results, correlating geometric phases with language properties and downstream monolingual and cross-lingual performance. Each of us will explore different properties that RankMe curves may be correlated with:

- **João**: Do multilingual LLMs exhibit synchronized geometric phases across languages during pretraining?
- **Pedro**: Do RankMe curves depend on language families or training data distribution?
- **Giacomo**: Do properties of RankMe evolution predict downstream monolingual performance?
- **Mattia**: Does the difference between curves predict cross-lingual transfer?

References

Project Apertus, Alejandro Hernández-Cano, Alexander Hägele, Allen Hao Huang, Angelika Romanou, Antoni-Joan Solergibert, Barna Pasztor, Bettina Messmer, Dhia Garbaya, Eduard Frank Ďurech, Ido Hakimi, Juan García Giraldo, Mete Ismayilzada, Negar Foroutan, Skander Moalla, Tiancheng Chen, Vinko Sabolčec, Yixuan Xu, Michael Aerni, Badr AlKhamissi, Inés Altemir Mariñas, Mohammad Hossein Amani, Matin Ansaripour, Ilia Badanin, Harold Benoit, Emanuela Boros, Nicholas Browning, Fabian Bösch, Maximilian Böther, Niklas Canova, Camille Challier, Clement Charmillot, Jonathan Coles, Jan Deriu, Arnout Devos, Lukas Drescher, Daniil Dzenhaliou, Maud Ehrmann, Dongyang Fan, Simin Fan, Silin Gao, Miguel Gila, María Grandury, Diba Hashemi, Alexander Hoyle, Jiaming Jiang, Mark Klein, Andrei Kucharavy, Anastasiia Kucherenko, Frederike Lübeck, Roman Machacek, Theofilos Manitaras, Andreas Marfurt, Kyle Matoba, Simon Matrenok, Henrique Mendonça, Fawzi Roberto Mohamed, Syrielle Montariol, Luca Mouchel, Sven Najem-Meyer, Jingwei Ni, Gennaro Oliva, Matteo Pagliardini, Elia Palme, Andrei Panferov, Léo Paoletti, Marco Passerini, Ivan Pavlov, Auguste Poiroux, Kaustubh Ponkshe, Nathan Ranchin, Javi Rando, Mathieu Sauser, Jakhongir Saydaliev, Muhammad Ali Sayfiddinov, Marian Schneider, Stefano Schuppli, Marco Scialanga, Andrei Semenov, Kumar Shridhar, Raghav Singhal, Anna Sotnikova, Alexander Sternfeld, Ayush Kumar Tarun, Paul Teiletche, Jannis Vamvas, Xiaozhe Yao, Hao Zhao, Alexander Ilic, Ana Klimovic, Andreas Krause, Caglar Gulcehre, David Rosenthal, Elliott Ash, Florian Tramèr, Joost VandeVondele, Livio Veraldi, Martin Rajman, Thomas Schulthess, Torsten Hoefler, Antoine Bosselut, Martin Jaggi, and Imanol Schlag. 2025. Apertus: Democratizing open and compliant llms for global language environments.

Terra Blevins, Hila Gonen, and Luke Zettlemoyer. 2022. Analyzing the mono- and cross-lingual pretraining dynamics of multilingual language models.

Tyler A. Chang, Zhuowen Tu, and Benjamin K. Bergen. 2022. The geometry of multilingual language model representations.

Alexis Conneau, Ruty Rinott, Guillaume Lample, Adina Williams, Samuel Bowman, Holger Schwenk, and Veselin Stoyanov. 2018. XNLI: Evaluating cross-lingual sentence representations. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2475–2485, Brussels, Belgium. Association for Computational Linguistics.

Quentin Garrido, Randall Balestriero, Laurent Najman, and Yann Lecun. 2023. Rankme: Assessing the downstream performance of pretrained self-supervised representations by their rank.

Omer Goldman, Uri Shaham, Dan Malkin, Sivan Eiger, Avinatan Hassidim, Yossi Matias, Joshua Maynez, Adi Mayrav Gilady, Jason Riesa, Shruti Rijhwani, Laura Rimell, Idan Szpektor, Reut Tsarfaty, and Matan Eyal. 2025. Eclektic: a novel challenge set for evaluation of cross-lingual knowledge transfer.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2020. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*.

Melody Zixuan Li, Kumar Krishna Agrawal, Arna Ghosh, Komal Kumar Teru, Adam Santoro, Guillaume Lajoie, and Blake A. Richards. 2025. Tracing the representation geometry of language models from pretraining to post-training.

Guilherme Penedo, Hynek Kydlíček, Vinko Sabolčec, Bettina Messmer, Negar Foroutan, Amir Hossein Kargaran, Colin Raffel, Martin Jaggi, Leandro Von Werra, and Thomas Wolf. 2025. Fineweb2: One pipeline to scale them all – adapting pre-training data processing to every language.

Edoardo M. Ponti, Goran Glavaš, Olga Majewska, Qianchu Liu, Ivan Vulić, and Anna Korhonen. 2020. XCOPA: A multilingual dataset for causal commonsense reasoning. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2362–2376. Association for Computational Linguistics.

Melissa Roemmele, Cosmin Adrian Bejan, and Andrew S Gordon. 2011. Choice of plausible alternatives: An evaluation of commonsense causal reasoning. In *2011 AAAI Spring Symposium Series*.

Haoran Sun, Renren Jin, Shaoyang Xu, Leiyu Pan, Supryadi, Menglong Cui, Jiangcun Du, Yikun Lei, Lei Yang, Ling Shi, Juesi Xiao, Shaolin Zhu, and Deyi Xiong. 2024. Fuxitranyu: A multilingual large language model trained with balanced data.

Hetong Wang, Pasquale Minervini, and Edoardo M. Ponti. 2024. Probing the emergence of cross-lingual alignment during llm training.